Fashion CBIR

Cascaded Deep and Classical Feature Fusion for Fashion Image Retrieval

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Motivation & Dataset

Kaggle Fashion Product Images

- Studio-lit catalog images of fashion products
- Annotated with masterCategory, subCategory, articleType, baseColour
- Images resized to 224 × 224 × 3 (RGB) for feature extraction
- **Evaluation: 200 random queries, Precision@K (K=1, 5, 10)**
- **Four hierarchy levels for precision measurement**
- Task: street-to-shop retrieval (query image → similar catalog items)

- Hybrid CBIR
- ResNet50 + HSV + LBP features
- Two-stage cascaded retrieval
- Late fusion (no retraining)

Key Challenge

The cross-domain gap between street photos (uncontrolled conditions) and studio catalog images makes naive pixel matching ineffective. Clothing deformation, varying angles, and occlusion require both semantic and fine-grained visual understanding.

Three Feature Descriptors

Semantic (CNN)

```
ResNet50 (ImageNet pretrained)
Global average pooling
2048-D embedding vector
```

Distance Metric

Cosine distance

2048-D output

Captures high-level semantics: garment type, silhouette, and overall style.

Color (HSV Histogram)

```
Convert to HSV color space
32 bins per channel (H, S, V)
Concatenate → 96-D vector
L2-normalized
```

Distance Metric

Euclidean distance

96-D output

HSV decouples chrominance from illumination, robust to lighting variation.

Texture (LBP)

```
Uniform LBP (P=24, R=3)
Histogrammed into 26 bins
L2-normalized
```

Distance Metric

Euclidean distance

26-D output

Encodes micro-texture patterns: fabric weave, surface smoothness, fine structure.

Technical Design & Infrastructure

Cascaded Retrieval Pipeline

Component	Detail
Stage 1 Filter	Cosine on CNN → Top 100
Stage 2 Re-rank	Euclidean on Color + LBP
Re-rank Score	d_color + 0.5 * d_texture
CNN Backbone	ResNet50 (ImageNet, no top, avg pool)
Index Storage	Separate .npy per descriptor + pkl
kNN Backend	sklearn NearestNeighbors (brute)
Feature Norm	L2-normalize all descriptors

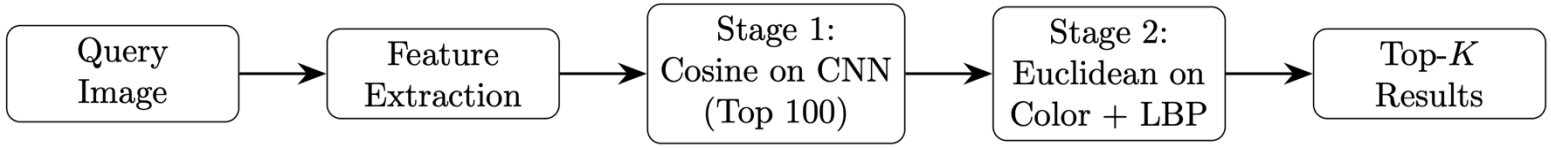
Infrastructure

- **CeSViMa Magerit Supercomputer**
- NVIDIA A100 (80 GB)
- SLURM job scheduler (sbatch)
- TensorFlow + OpenCV + scikit-image
- 4 CPUs, 80 GB RAM per job

Ablation Study

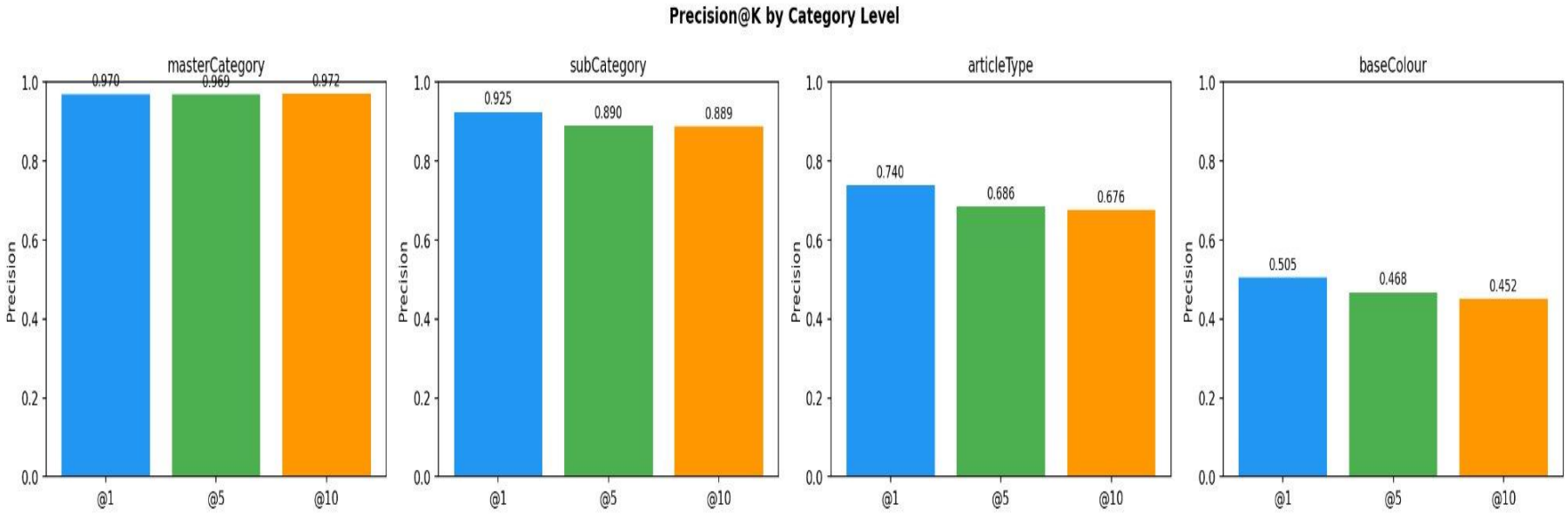
Baseline 1: ResNet-only (cosine on CNN)
Baseline 2: Color-only (Euclidean on HSV)
Full system: Cascaded (CNN filter + re-rank)
Metric: Precision@K at 4 metadata levels
(200 random queries per strategy)

Two-Stage Cascaded Retrieval Pipeline



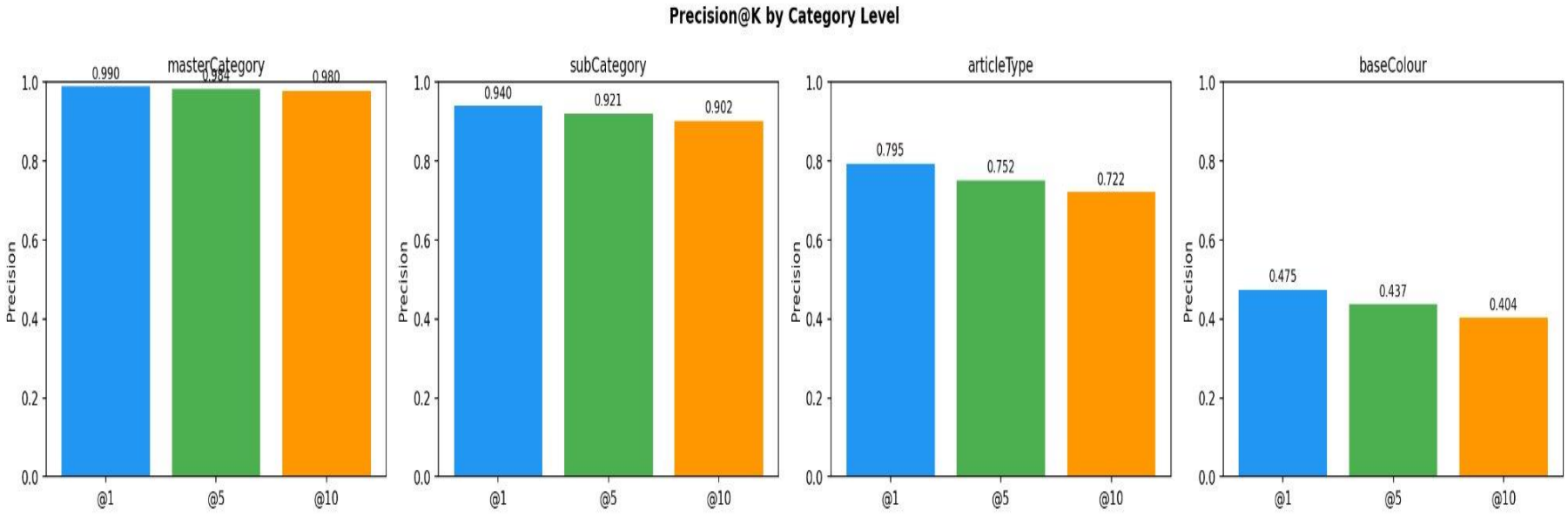
Design Rationale: Late fusion design: Stage 1 uses cosine distance on ResNet50 embeddings to filter the full 44K index down to 100 semantically similar candidates. Stage 2 re-ranks these 100 by Euclidean distance on HSV color + LBP texture descriptors (d_color + 0.5 × d_texture). Each metric is matched to its descriptor type cosine for high-dimensional CNN, Euclidean for compact histograms.

Precision@K: Cascaded Strategy



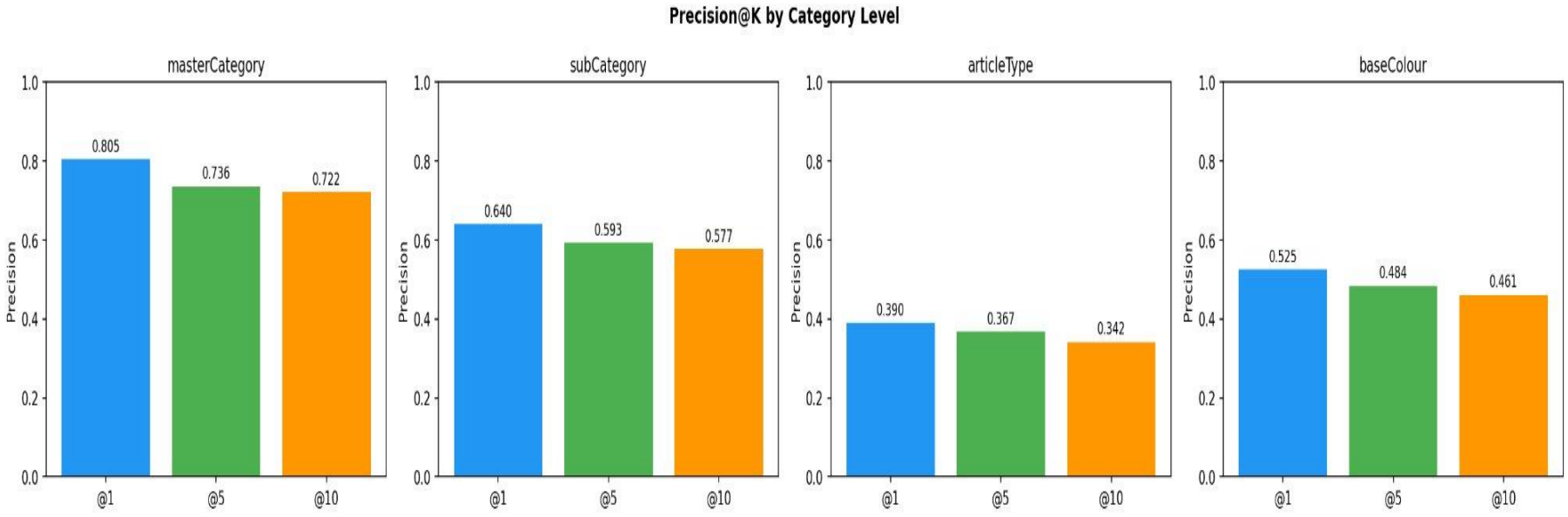
Key Observations: Cascaded strategy (44,441 indexed images, 200 queries). masterCategory near-perfect (~97%) CNN semantics alone resolve broad categories. subCategory strong (~89%) with slight drop at deeper K. articleType drops to ~68–74%, finer garment types are harder to distinguish. baseColour is the hardest level (~45–50%), showing that color matching remains challenging even with HSV re-ranking.

Precision@K: ResNet-Only Strategy



Key Observations: ResNet-only baseline (cosine on CNN embeddings). masterCategory reaches ~98–99%, deep features almost perfectly resolve broad categories. subCategory ~90–94% and articleType ~72–79%, both higher than cascaded, because cosine on CNN is already a strong semantic matcher. However, baseColour drops to ~40–47%, well below both cascaded (~45–50%) and color-only (~46–52%), confirming that CNN embeddings encode shape/category but not fine color differences.

Precision@K: Color-Only Strategy

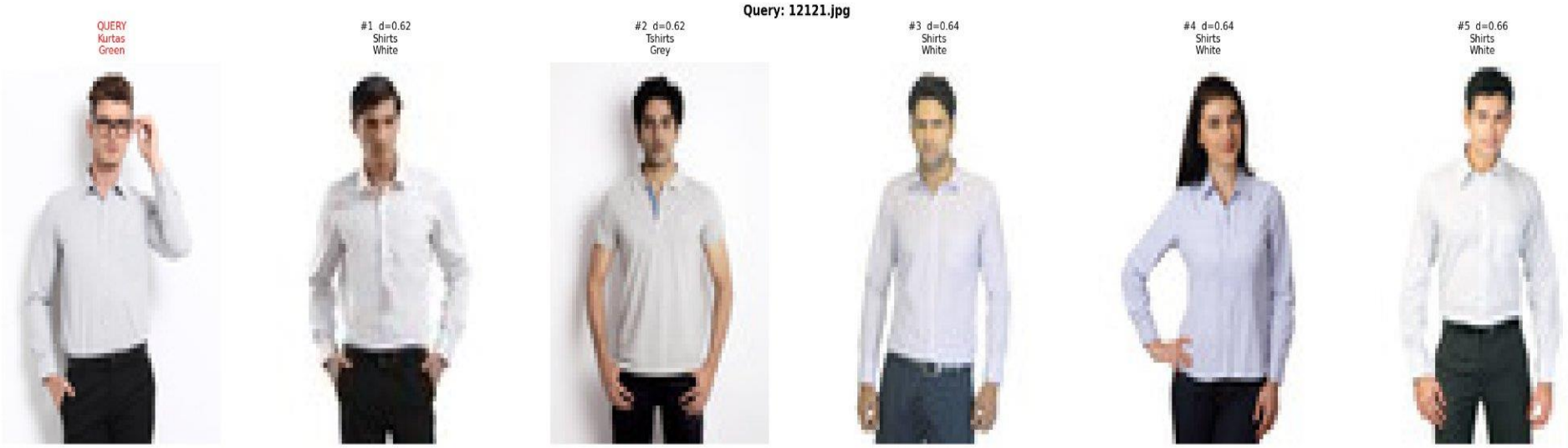


Key Observations: Color-only baseline (Euclidean on HSV histograms). masterCategory drops to ~72–80%. color alone partially resolves broad categories but confuses items sharing similar palettes. subCategory ~58–64%. articleType collapses to ~34–39%, confirming that color cannot distinguish garment types. Interestingly, baseColour (~46–52%) is comparable to cascaded, since HSV histograms directly encode the attribute being measured.

Demo: Handbags: Color Re-ranking in Action

Observations: Query: orange handbag. All 5 results are handbags (100% articleType precision). Stage 1 CNN filtering correctly narrows to bags. Stage 2 HSV re-ranking produces a smooth color gradient: orange → orange → peach → brown → red, showing the cascaded pipeline sorting by hue similarity. Distances increase gradually (0.17 → 0.25).

Demo: Kurtas: Cross-Category Confusion



Observations: Query: green kurta. All 5 results are Shirts or Tshirts: 0% articleType match. The CNN treats kurtas and shirts as visually similar (both upper-body garments on models with similar poses). High distances (0.62–0.66) signal low confidence. This illustrates the limit of transfer learning: ResNet50 was trained on ImageNet, not fashion-specific categories.

Results & Conclusions

Strategy	P@1	P@5	P@10
ResNet-only (cosine)	79.5%	75.2%	72.2%
Color-only (Euclidean)	39.0%	36.7%	34.2%
Cascaded (CNN + Color + LBP)	**74.0%**	**68.6%**	**67.6%**

Best Strategy (articleType level): ResNet-only leads at articleType (P@1 = 79.5%). Cascaded adds +5% on baseColour over ResNet. Color-only confirms that semantics, not color, drive retrieval.

Key Findings

- **ResNet-only is the strongest single strategy at semantic levels**

P@1 = 79.5% at articleType vs. cascaded 74.0%. CNN embeddings capture garment type, silhouette, and style.

- **Cascaded adds value at the baseColour level**

P@1 = 50.5% vs. ResNet-only 47.5%. HSV re-ranking adds +3% by matching exact shades after semantic filtering.

- **Color-only alone is insufficient for retrieval**

articleType P@1 = 39.0%, color cannot distinguish garment types. But baseColour (52.5%) rivals cascaded, validating HSV as a color-specific descriptor.